MICRON LABORATORY SERVICES

sanas Calibration Laboratory
CAL 103-03-00
CAL 103-14-00

9 Beatty Avenue, Witbank 1035 Tel: 013 690 1532 Fax: 013 656 1000
Postnet Suite 139 Private Bag X 7260 Witbank 1035
Reg No: 2017/365305/07 Vat nr: 4430102030
www.micronlab.co.za

CERTIFICATE OF CALIBRATION

Certificate no:	AZU AF1.2
Date of calibration:	27.02.2026
Date of next calibration:	March 2027
Customer:	AzureSky (Pty) Ltd Unit 3, 17 Markiel Road Wadeville Germiston, 1401
Location:	Laboratory
Contact:	nonhlanhla@azuresky.it.com

Conditions of Issue:

This certificate is issued in accordance with the conditions of accreditation granted by the South African National Accreditation System of South Africa. Subject to the uncertainties specified, this is a correct record of measurements made.

Copyright of this certificate is owned jointly by SANAS and the issuing laboratory. This certificate may not be reproduced other than in full without written approval of SANAS and of the issuing laboratory.

The values in this certificate were correct at the time of calibration. Subsequently the accuracy will depend on such factors as, the care of handling, the use of the instrument and the frequency of use. Re-calibrations should be performed after a period which has been chosen to ensure that the instrument's accuracy remains within the desired limits.

Where statements of conformity are made in the certificate, the following rules are applied:
Within Tolerance: Results ± expanded uncertainty within limits / specification.
Out of Tolerance: Results ± expanded uncertainty exceeds limits / specification.
Indeterminate: Results ± expanded uncertainty overlaps limits / specification.

Measurement results relate only to the item calibrated.

Technical Signatory: M Stroebel

Calibrated by: F Rossouw

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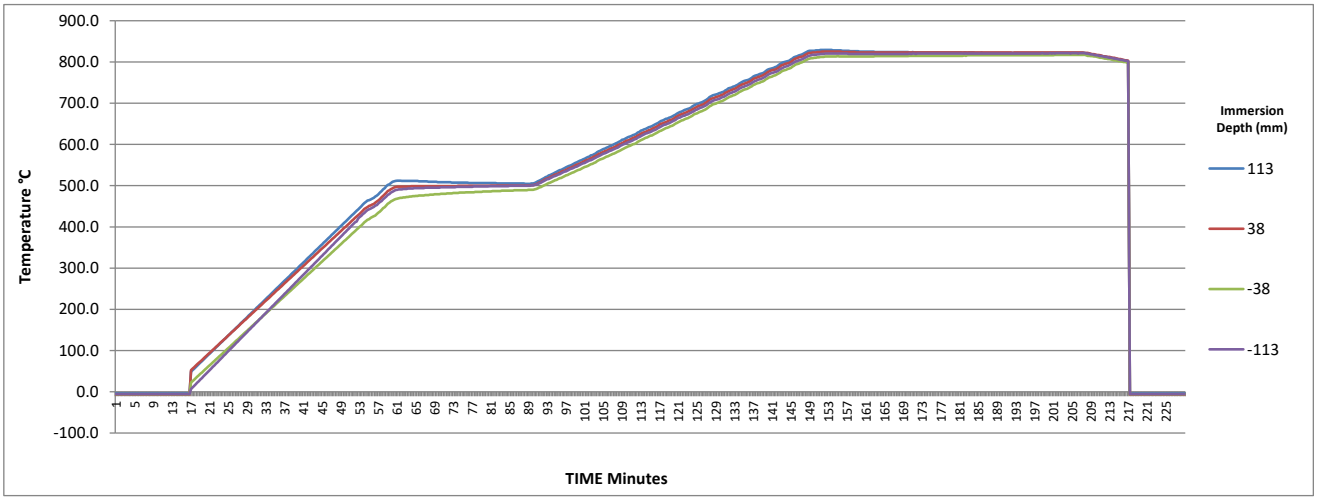
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CERTIFICATE OF CALIBRATION

Customer:	AzureSky (Pty) Ltd	**Certificate Number:**	AZU AF1.2
Contact:	nonhlanhla@azuresky.it.com	**Instrument:**	ATS Ashing Furnace
Location:	Laboratory	**Serial Number:**	ATS/210325/74A
Date of Calibration:	27.02.2026	**Company Number:**	N.N
Date of Next Calibration:	March 2027	**Procedure Reference:**	P4.1.3.ISO ASH

Calibration Standards	**Description**	**Model**	**Coy No**	**Re-calibration date**
	OMEGA LOGGER	HH309A (V 4.0.1.0)	A-DL14	31st October 2026

Calibration Results - °C



Uncertainty of Measurement ± °C	4.0

Method: The device under test (DUT) was calibrated by comparison to the laboratory standard at the listed temperature, temperature readings were taken over a period to include deviation from set point.
Uncertainty: The reported uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor K=2. Which for a normal distribution corresponds to a probobility of approximatley 95%. The uncertainty in the calculated uncertainty of measurement for all of the individual readings

Traceability: The standards used for calibration are traceable to the national measuring standard.

Average Set Temperature At 815°C Set Point	**820**	**Average Set Temperature At 500°C Set Point**	**492**
Maximun Temperature At 815°C Set Point	**823**	**Maximun Temperature At 500°C Set Point**	**503**
Minimum Temperature At 815°C Set Point	**815**	**Minimum Temperature At 500°C Set Point**	**479**
Standard Deviation At 815°C Set Point	**0.5**	**Standard Deviation At 500°C Set Point**	**2.1**

Remarks	

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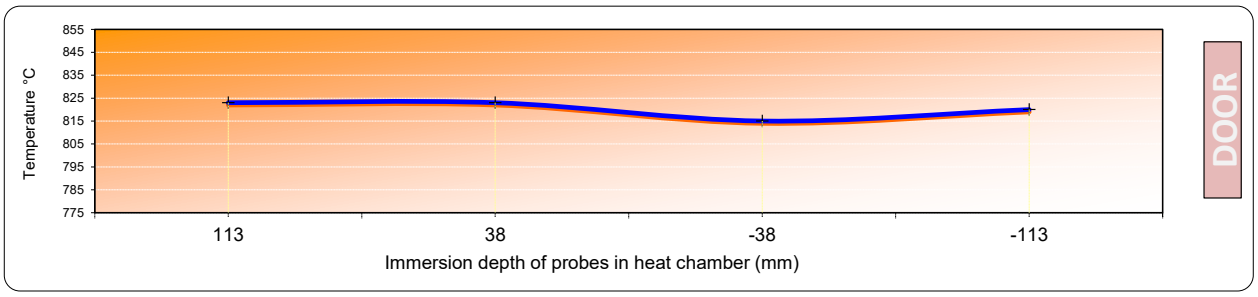
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Calibration Standards	**Description**	**Model**	**Coy No**	**Re-calibration date**
	OMEGA LOGGER	HH309A (V 4.0.1.0)	A-DL14	31st October 2026

Calibration Results - °C



0 mm

225 mm

Nominal Value (°C)	Immersion From Door (mm)	Before (°C)	After (°C)	Average Temperature (°C)
815	113	822	823	820
	38	822	823	
	-38	814	815	
	-113	819	820	

Uncertainty Of Measurement °C ± 4	The above readings are the average taken at 20 second intervals for a period of 60 minutes

Method: The device under test (DUT) was calibrated by comparison to the laboratory standard at the listed temperature, temperature readings were taken over a period to include deviation from set point.

Uncertainty: The reported uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor K=2. Which for a normal distribution corresponds to a probobility of approximatley 95%. The uncertainty in the calculated uncertainty of measurement for all of the individual readings.

Traceability: The standards used for calibration are traceable to the national measuring standard.

Controller Settings	**Controller Off-Set °C**	-1	
	Proportional Band (P)	0.9	**Note:** Adjustment to any of these controller settings will affect the unit accuracy, ramp rate and STD deviation.
	Integration Time (I)	230	
	Derivative Time (D)	58	
Laboratory enviroment during calibration	**Temperature at start °C**	26	± 2°C
	Temperature at end °C	28	± 2°C
Remarks			

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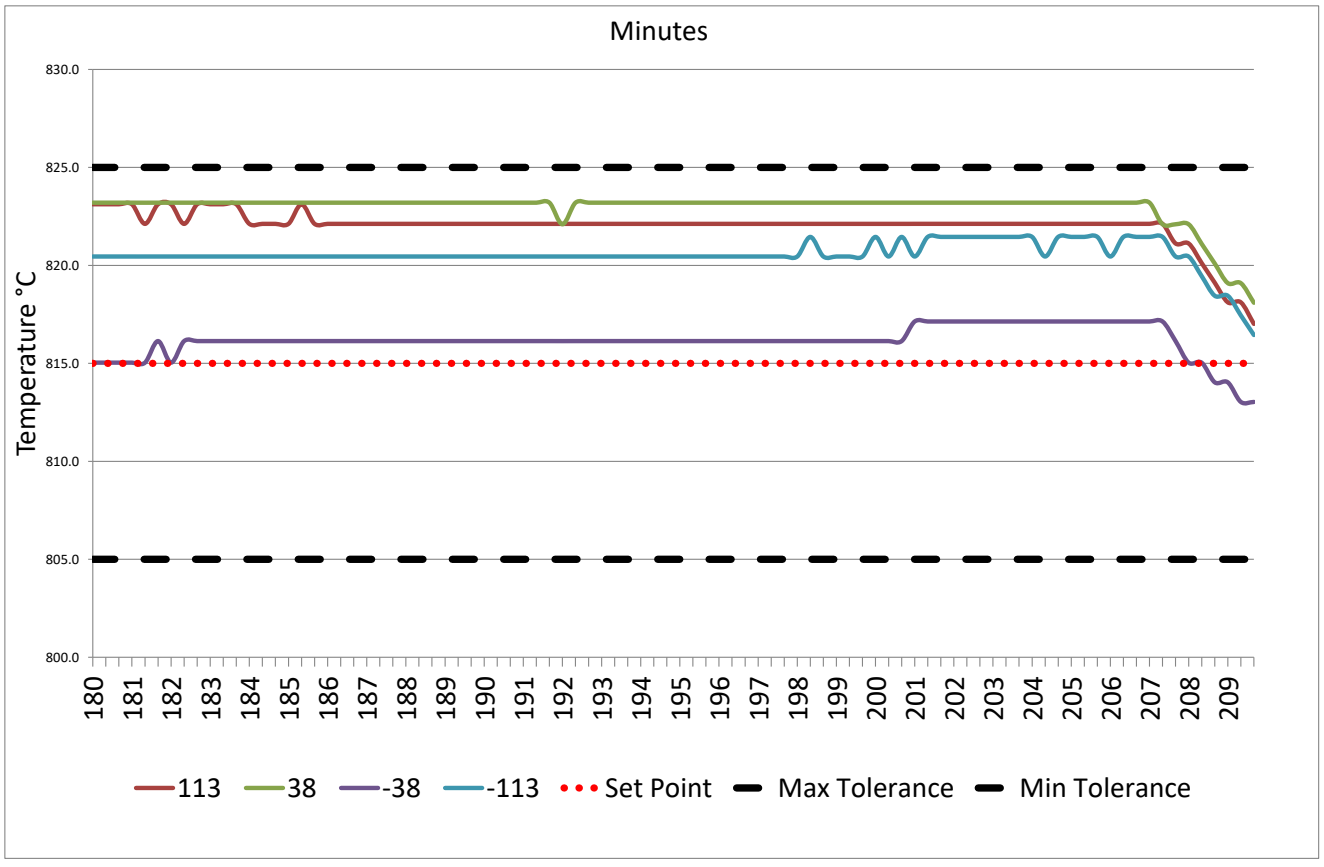
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Date of Next Calibration:	March 2027	**Procedure Reference:**	P4.1.3.ISO ASH

Calibration Standards	**Description**	**Model**	**Coy No**	**Expiry Date**
	OMEGA LOGGER	HH309A (V 4.0.1.0)	A-DL14	31st October 2026

TEMPERATURE MAPPING AT SET POINT



Technical Signatory: M Stroebel

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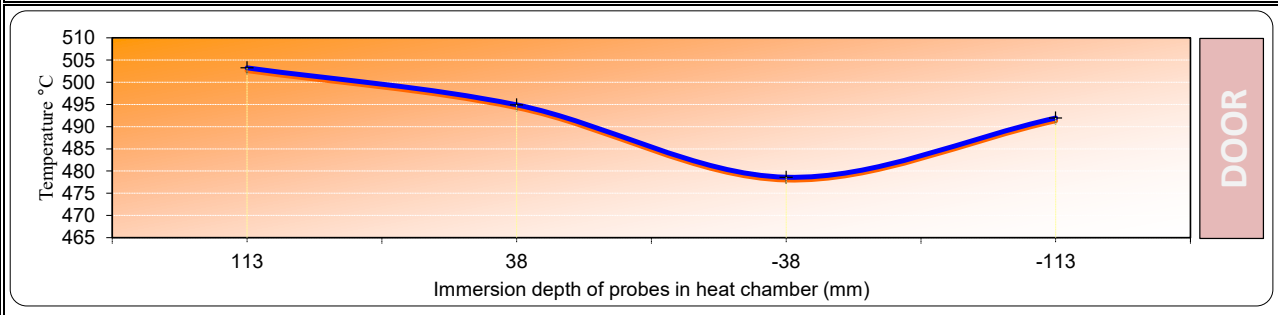
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Calibration Standards	**Description**	**Model**	**Coy No**	**Re-calibration date**
	OMEGA LOGGER	HH309A (V 4.0.1.0)	A-DL14	31st October 2026

CALIBRATION RESULTS - ºC



Nominal Value (°C)	Immersion From Door (mm)	Before (°C)	After (°C)	Average Temperature (°C)
500	113	503	503	492
	38	494	495	
	-38	478	479	
	-113	491	492	

Uncertainty Of Measurement °C ±	5	The above readings are the average taken at 20 second intervals for a period of 30 minutes

Method: The device under test (DUT) was calibrated by comparison to the laboratory standard at the listed temperature, temperature readings were taken over a period to include deviation from set point.
Uncertainty: The reported uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor K=2. Which for a normal distribution corresponds to a probobility of approximatley 95%. The uncertaintv in the calculated uncertaintv of measurement for all of the individual readinas.
Traceability: The standards used for calibration are traceable to the national measuring standard.

Controller Settings	**Controller Off-Set**	**°C**	-1	**Note:** Adjustment to any of these controller settings will affect the unit accuracy, ramp rate and STD deviation.
	Proportional Band	**(P)**	1.3	
	Integration Time	**(I)**	270	
	Derivative Time	**(D)**	67	

Remarks	

Technical Signatory: M Stroebel

Calibrated by: F Rossouw

End of Certificate